CADILLAC
1877
TREE CITY
MICHIGAN

5

CITY OF CADILLAC WWTP
2003 - EPA BIOSOLIDS REPORT

The City of Cadillac WWTP is an advanced treatment plant providing treatment to the City of Cadillac and to several surrounding townships. The daily flow for 2003 averaged 2.085 MGD. The treatment process is comprised of grit removal, primary treatment, secondary treatment (activated sludge), RBC treatment for ammonia removal, sand filtration, chlorination and dechlorination. The effluent is discharged into the Clam River which is designated as a cold water trout stream several hundred yards from our discharge.

All solids are removed from the primary tanks to the anaerobic digester . From the primary digester solids are transferred to a secondary covered digester which is not heated. this tank is used for solids separation and supernate withdrawal. Solids from this tank is pumped to one of two storage tanks. Two times a year the city contracts with a sludge application firm for solids injection on farm lands.

The following Biosolids Report is comprised of the DEQ BIOSOLIDS ANNUAL REPORT mailed to my attention, information on pathogen reduction, vector attraction reduction, sludge unit process operation and laboratory analysis. Also attached are the required certification statements. If you need any additional information, please feel free to contact myself at the following location :

266 DT
112 A

Jeffrey M. Dietlin
Cadillac Utilities Department
200 N. Lake Street
Cadillac, MI 49601
Phone: 231/775-2841
Fax: 231/775-2876

CADILLAC WWTP - 2003 EPA SLUDGE REPORT

Pathogen Reduction Requirements

The Cadillac WWTP uses the anaerobic sludge digestion process. All primary and waste activated sludge is pumped into the primary digester. The primary digester is heated to 97.9 F using an external heat exchanger. The primary digester also has a PERTH gas recirculation system for complete mixing. The current retention time is approximately 20.0 days. The digested sludge is then pumped to a secondary anaerobic digested which is non-heated and used for supernate draw-off.

The digested sludge from the secondary digester is then pumped into a sludge holding tank for storage until land application in the spring and fall.

Vector Attraction Reduction Requirements

The Cadillac WWTP uses volatile solids reduction as a method to reduce vector attraction and meet Class B requirements. The attached worksheets indicates that we had an average of 52.61% volatile solids reduction per EPA manual, EPA 831B-93-002a, instructions. We also sub-surface inject our Biosolids.

Management Practices

The Cadillac WWTP contracted with BioTech Agronomics for land application. The WWTP finds all the land and is responsible for soil testing. The contractor is responsible for soil injection of the digested sludge. The contractor uses 9000 gallon nurse trucks to deliver sludge to the fields and a 4000 gallon "Big A" land applicator for the injection.

All sludge is applied to private farmland for corn silage for cattle. The sludge is applied in either the spring or fall prior to planting and the crop is not removed until the following fall. All land is private with no public access to the fields.

(1) Cadillac Dist

DEQ

Michigan Department Of Environmental Quality - Water Division

BIOSOLIDS ANNUAL REPORT

SECTION I – BIOSOLIDS LAND APPLICATION REPORT

(5)

By Authority of Part 31, Water Resources Protection, of 1994 PA 451, as amended, (Part 31), this form is to be used by generators and distributors to report biosolids applied to the land which are subject to Part 31. Failure to properly report this information is a violation of Act 451 and is subject to penalties as provided. The information provided herein will be used to determine fees to support the program in accordance with Part 31.

REPORTS ARE DUE OCTOBER 30, 2003

Please note: All Treatment Works Treating Domestic Sewage (TWTDS) are required to complete and return this form.

**** If you did not land apply please put 0 for the tons land applied and return only this page to the address below.**
**** If you landfilled your biosolids list the tons that were landfilled and return only this page to the address below.**
**** If you incinerated any portion of your biosolids you must still attach the appropriate DMR's.**
**** If you hauled liquid biosolids to another facility, list the amount hauled and the haulers name.**

REQUIRED INFORMATION - TO BE COMPLETED BY GENERATOR OR DISTRIBUTOR. (Please type or print.)

FACILITY NAME City of Cadillac			NPDES, State, or COC Permit Number MI0020257 ✓
FACILITY ADDRESS 1121 Plett Road			TELEPHONE NO. 231-775-2841
CITY Cadillac	STATE MI	ZIP 49601	CONTACT PERSON Jeffrey M. Dietlin

DURING FISCAL YEAR 2003 (10/1/2002 - 9/30/2003) , THE GENERATOR/DISTRIBUATOR NAMED ABOVE LAND APPLIED

266 ✓ DRY TONS OF BIOSOLIDS — 241.2 DRY METRIC TONS OF BIOSOLIDS TO LANDS WITHIN THE STATE OF MICHIGAN

266 TOTAL DRY TONS OF BIOSOLIDS GENERATED

0 TOTAL DRY TONS LANDFILLED

0 TOTAL DRY TONS INCINERATED

0 TOTAL DRY TONS TRANSPORTED OUT OF STATE

0 TOTAL GALLONS TRANSPORTED TO ANOTHER WASTEWATER TREATMENT FACILITY

_______________ RECEIVING FACILITY NAME

_______________ HAULERS NAME

To convert the English system (short tons) to metric tons, use the following equation: DRY METRIC TONS = DRY SHORT TONS x .907

I certify that the information as provided on this form is true.

[signature] 10/29/2003

Signature of Authorized Representative — Date

REQUIRED INFORMATION. COMPLETE TO ENSURE YOU RECEIVE YOUR INVOICE IN A TIMELY MANNER.

MAILING NAME City of Cadillac WWTP			
MAILING ADDRESS 200 N. Lake Street			
MAILING CITY Cadillac	STATE Mi	ZIP 49601	CONTACT PERSON Jeffrey M. Dietlin

IF YOU HAVE ANY QUESTIONS ABOUT COMPLETING THIS FORM, PLEASE CONTACT THE DEQ DISTRICT STAFF PERSON FOR YOUR AREA

PLEASE RETURN COMPLETED FORM TO:

BIOSOLIDS PROGRAM
TECHNICAL SUPPORT UNIT
WATER DIVISION
DEPARTMENT OF ENVIRONMENTAL QUALITY
PO BOX 30273
LANSING MI 48909-7773

EQP 5865 (Rev. 7/03)

DEQ

Michigan Department of Environmental Quality – Water Division

BIOSOLIDS ANNUAL REPORT

SECTION II – GENERAL FACILITY INFORMATION

By Authority of Part 31, Water Resources Protection of 1994 PA 451, as amended, (Part 31), these forms are to be used by generators and distributors to report biosolids applied to the land which are subject to Part 31. Failure to properly report this information is a violation of Act 451 and is subject to penalties as provided.

1. Annual Reporting Year October 1, 2002 to September 30, 2003	2. NPDES or COC Number **MI002057**
3. Generator Name **CADILLAC WWTP**	4. Facility Name (if Different)
5. Latitude (nearest 15 seconds) **N44 15.915** — Longitude **W085 23.851**	6. Plant Type **ACTIVATED SLUDGE TERTIARY TREATMENT**
7. Permit Issued (Date) **3/10/1998 (NEW PERMIT NOT ISSUED YET)**	8. Permit Expires (Date) **10/01/2002**
9. Current Actual Flow Rate (MGD) **2.085**	10. Industrial Pretreatment? (check one) ☒ YES ☐ NO
11. Facility sends biosolids out of state? (Y/N) ☐ YES ☒ NO	

12. Facility Physical Address

Street: **1121 PLETT ROAD**		City: **CADILLAC**
County: **WEXFORD**	Zip Code: **49601**	Phone (include area code): **231-775-2841**

13. Facility Mailing Address (if different)

Street: **200 N. LAKE STREET**		City: **CADILLAC**
County: **WEXFORD**	Zip Code: **49601**	Phone (include area code): **231-775-2841**

14. Name of Responsible Official **Jeffrey M. Dietlin**	15. Title of Responsible Official **Water Resources Supervisor**

16. Facility Contact Person Information

Name of Contact **Jeffrey M. Dietlin**	Title **Water Resources Supervisor**	
E-Mail Address **wwtpr@cadillac-mi.net**	Phone **231-775-2841**	Fax **231-775-2876**

17. Contract Applier(s)/Hauler(s) Information

Name of Contractor **BioTech Agronomics**	
Phone **231-325-5011**	Contact **Kevin Bonney**
Name of Contractor	
Phone	Contact

Please place all attachments at the end of the report packet as appendices <u>not</u> after each section

EQP 5865 (7/2003)

DEQ

Michigan Department of Environmental Quality – Water Division

(1) Cadillac Dist

BIOSOLIDS ANNUAL REPORT

SECTION V – MONITORING DATA SUMMARY

(5) City of Cadillac

Parameter	Minimum Monthly Concentration	Average Monthly Concentration	Maximum Monthly Concentration	Units	# of Analyses	Method Detection Limit	Test Method	Sample Type
Inorganics								
Total Solids	5.0	5.16	5.33	%	2	0.01	Sm(18) 2540B	☐ Grab ☒ Composite
Total Arsenic	20.3	20.6	21	Mg/kg	2	4	EPA 6020	☐ Grab ☒ Composite
Total Cadmium	2.1	2.8	3.6	Mg/kg	2	1	EPA 6020	☐ Grab ☒ Composite
Total Copper	903	914	925	Mg/kg	2	15	EPA 6010B	☐ Grab ☒ Composite
Total Lead	61	72	82	Mg/kg	2	8	EPA 6010B	☐ Grab ☒ Composite
Total Mercury	2.59	4.3	6	Mg/kg	2	1.7	EPA 7471A	☐ Grab ☒ Composite
Total Molybdenum	37	37	37	Mg/kg	2	10	EPA 6010B	☐ Grab ☒ Composite
Total Nickel	28	30	31	Mg/kg	2	10	EPA 6010B	☐ Grab ☒ Composite
Total Selenium	4.3	7.2	<10	Mg/kg	2	10	EPA 6020	☐ Grab ☒ Composite
Total Zinc	1430	1565	1700	Mg/kg	2	15	EPA 6010B	☐ Grab ☒ Composite
Nutrients								
Total Kjeldahl Nitrogen	44300	46850	49400	Mg/kg	2	2000	EPA 351.2	☐ Grab ☒ Composite
Ammonium Nitrogen	18600	19100	19600	Mg/kg	2	400	EPA 350.1	☐ Grab ☒ Composite
Total Phosphorus	25000	34200	43400	Mg/kg	2	800	EPA 365.2	☐ Grab ☒ Composite
Total Potassium	1510	1835	2160	Mg/kg	2	66	EPA 6010B	☐ Grab ☒ Composite

Include copies of the actual analytical laboratory data sheets as an attachment at the end of the packet. All sampling shall be representative of the biosolids applied to land during the reporting period and in accordance with R 323.2415 (2) Frequency of Monitoring – Land Application. Analytical methods shall be in accordance with R 323.2406 (2) Methods for Biosolids. All analysis should be provided on a dry weight basis.

If you have any questions about the preparation of this form, contact the DEQ district biosolids program staff person for your area.

EQP 5885 (7/2003)

DEQ

Michigan Department of Environmental Quality – Water Division

BIOSOLIDS ANNUAL REPORT

SECTION III – FINAL USE/DISPOSAL PRACTICES (reporting year 2003)

1. Land Application (total) 266 dt

Bulk Biosolids:	266 dt	Derived Materials:	0 dt
Agricultural Land	266 dt	Agricultural Land	dt
Forest	dt	Forest	dt
Public Contact Site	dt	Public Contact Site	dt
Reclamation Site	dt	Reclamation Site	dt
Sold or Given Away	dt	Sold or Given Away	dt
Lawn or Garden	dt	Lawn or Garden	dt

2. Surface Disposal (Total) 0 dt
   - With Liner and LCS dt
   - Without Liner and LCS dt

3. Landfill (Total) 0 dt
   - Landfill Disposal dt
   - Landfill Cover dt
   - Landfill Name

4. Incineration 0 dt

5. Transported to Another Facility 0 dt
   - Name
   - Address
   - NPDES
   - Phone

6. Received From Another Facility 0 dt
   - Name
   - Address
   - NPDES
   - Phone

7. Other 0 dt

8. Stored 0 dt

9. Certifications: **(*Please Attach All Required Certification Statements)**

Pathogen Certification (select one)	☒ YES	☐ NO	☐ NOT APPLICABLE
Vector/Attraction Certification? (select one)	☒ YES	☐ NO	☐ NOT APPLICABLE
Management Practice Certification? (select one)	☒ YES	☐ NO	☐ NOT APPLICABLE
CPLR Certification? (select one)	☐ YES	☐ NO	☒ NOT APPLICABLE
- CPLR Site Restrictions Certification? (select one)	☐ YES	☐ NO	☒ NOT APPLICABLE

**dt = English Dry Tons
**CPLR: Cumulative Pollutant Loading Rate – when pollutants exceed Table 3 concentrations (mg/kg)

If you have any questions about the preparation of this form, contact the DEQ district biosolids program staff person for your area.

EQP 5865 (7/2003)

DEQ

Michigan Department of Environmental Quality – Water Division

BIOSOLIDS ANNUAL REPORT

SECTION IV – LAND APPLICATION SITE INFORMATION

SITE GB001 - INFORMATION

Site Name GB001	Site Number	Indian Country ☐ YES ☒ NO
Owner Gerald Bilyea		
Operator City of Cadillac		
Applier BioTech Agronomics		
Latitude N44 17.607	Longitude W085 13.051	Reached 90% CPLR App. Rate? ☐ YES ☒ NO
Township Lake	Range 08W	Section 24
Acres 40	Acres Used 36	Crop Corn
Application Rate (tons/acre) 2.5	Notification (select one) ☒ YES ☐ NO	Cumulative Load Required (select one) ☐ YES ☒ NO

SITE GB002 - INFORMATION

Site Name GB002	Site Number	Indian Country ☐ YES ☒ NO
Owner Gerald Bilyea		
Operator City of Cadillac		
Applier BioTech Agronomics		
Latitude N44 17.826	Longitude W085 13.092	Reached 90% CPLR App. Rate? ☐ YES ☒ NO
Township Lake	Range 08W	Section 13
Acres 40	Acres Used 36	Crop corn
Application Rate (tons/acre) 2.45	Notification (select one) ☒ YES ☐ NO	Cumulative Load Required (select one) ☐ YES ☒ NO

SITE WE001 – INFORMATION

Site Name WE001	Site Number	Indian Country ☐ YES ☒ NO
Owner William Eubank		
Operator City of Cadillac		
Applier BioTech Agronomics		
Latitude N44 17.340	Longitude W085 13.173	Reached 90% CPLR App. Rate? ☐ YES ☒ NO
Township Lake	Range 08W	Section 24
Acres 40	Acres Used 40	Crop corn
Application Rate (tons/acre) 2.2	Notification (select one) ☒ YES ☐ NO	Cumulative Load Required (select one) ☐ YES ☒ NO

****Attach additional copies of this sheet as necessary, or you may attach your contractor's Land Application Reports, or use the DEQ Biosolids Recycling Sheet**

If you have any questions about the preparation of this form, contact the DEQ district biosolids program staff person for your area.

EQP 5865 (7/2003)

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DEQ

Michigan Department of Environmental Quality – Water Division

(1) Cadillac Dist

BIOSOLIDS ANNUAL REPORT

SECTION V – MONITORING DATA SUMMARY

(5) City of Cadillac

M10020257

Parameter	Minimum Monthly Concentration	Average Monthly Concentration	Maximum Monthly Concentration	Units	# of Analyses	Method Detection Limit	Test Method	Sample Type
Inorganics								
Total Solids	6.0	6.16	6.33	%	2	0.01	Sm(18) 2540B	☐ Grab ☒ Composite
Total Arsenic	20.3	20.6	21	Mg/kg	2	4	EPA 6020	☐ Grab ☒ Composite
Total Cadmium	2.1	2.8	3.6	Mg/kg	2	1	EPA 6020	☐ Grab ☒ Composite
Total Copper	903	914	925	Mg/kg	2	15	EPA 6010B	☐ Grab ☒ Composite
Total Lead	61	72	82	Mg/kg	2	8	EPA 6010B	☐ Grab ☒ Composite
Total Mercury	2.69	4.3	6	Mg/kg	2	1.7	EPA 7471A	☐ Grab ☒ Composite
Total Molybdenum	37	37	37	Mg/kg	2	10	EPA 6010B	☐ Grab ☒ Composite
Total Nickel	28	30	31	Mg/kg	2	10	EPA 6010B	☐ Grab ☒ Composite
Total Selenium	4.3	7.2	<10	Mg/kg	2	10	EPA 6020	☐ Grab ☒ Composite
Total Zinc	1430	1565	1700	Mg/kg	2	16	EPA 6010B	☐ Grab ☒ Composite
Nutrients								
Total Kjeldahl Nitrogen	44300	46850	49400	Mg/kg	2	2000	EPA 351.2	☐ Grab ☒ Composite
Ammonium Nitrogen	18600	19100	19600	Mg/kg	2	400	EPA 350.1	☐ Grab ☒ Composite
Total Phosphorus	25000	34200	43400	Mg/kg	2	800	EPA 365.2	☐ Grab ☒ Composite
Total Potassium	1510	1835	2160	Mg/kg	2	66	EPA 6010B	☐ Grab ☒ Composite

<u>Include copies of the actual analytical laboratory data sheets as an attachment at the end of the packet.</u> All sampling shall be representative of the biosolids applied to land during the reporting period and in accordance with R 323.2415 (2) Frequency of Monitoring – Land Application. Analytical methods shall be in accordance with R 323.2406 (2) Methods for Biosolids. All analysis should be provided on a dry weight basis.

If you have any questions about the preparation of this form, contact the DEQ district biosolids program staff person for your area.

EQP 5865 (7/2003)

DEQ

Michigan Department of Environmental Quality – Water Division

BIOSOLIDS ANNUAL REPORT

SECTION VI – PATHOGEN AND VECTOR ATTRACTION REDUCTION

1. Pathogen Reduction
Class A

- ☐ Class A – Alternative 1 (+ elevated temp for specified time)
- ☐ Class A – Alternative 2 (+ pH adjust for specified time/temp)
- ☐ Class A – Alternative 3 (+ virus and helminth criteria)
- ☐ Class A – Alternative 4 (+ other virus and helminth criteria)
- ☐ Class A – Alternative 5 (indicate which PFRP)
  - ☐ (a) composting
  - ☐ (b) heat drying
  - ☐ (c) heat treatment
  - ☐ (d) thermophillic aerobic digestion
  - ☐ (e) beta ray irradiation
  - ☐ (f) gamma ray irradiation
  - ☐ (g) pasteurization
- ☐ Class A – Alternative 6 (attach PFRP equivalent documentation)

2. Pathogen Reduction
Class B

- ☐ Class B – Alternative 1 (geometric mean of 7 samples)
- ☒ Class B – Alternative 2 (indicate which PSRP)
  - ☐ (a) aerobic digestion
  - ☐ (b) air drying
  - ☒ (c) anaerobic digestion
  - ☐ (d) composting
  - ☐ (e) lime stabilization (pH at 25' C or equivalent)
- ☐ Class B – Alternative 3 (attach PSRP equivalent documentation)

3. Vector Attraction Reduction
Method Used:

- ☒ Option 1 (minimum 38 percent reduction in volatile solids)
- ☐ Option 2 (Anaerobic process, with bench-scale demonstration)
- ☐ Option 3 (Aerobic Process, with bench scale demonstration)
- ☐ Option 4 (Specific Oxygen Uptake Rate (SOUR), aerobically digested)
- ☐ Option 5 (Aerobic Process plus raised temperature)
- ☐ Option 6 (Raise pH to 12 and retain at 11.5)
- ☐ Option 7 (75% solids with no unstabilized solids)
- ☐ Option 8 (90% solids with unstabilized solids)
- ☒ Option 9 (Injection below land surface with significant soil coverage)
- ☐ Option 10 (Covering active sewage sludge unit daily)

****Attach all Pathogen Reduction and Vector Attraction Reduction documentation to demonstrate compliance at the end of the packet**

If you have any questions regarding the preparation of this form, contact the DEQ district biosolids program staff person for your area.

EQP 5865 (7/2003)

DEQ

Michigan Department of Environmental Quality – Water Division

BIOSOLIDS ANNUAL REPORT

SECTION VII – SIGNATURE PAGE

Facility Name **Cadillac Waste Water Treatment Plant**	NPDES or COC Permit Number **MI0020257**

CERTIFICATION

"I certify under penalty of law that this document and all attachments were prepared under my direction or supervision in accordance with the system designed to assure that qualified personnel properly gather and evaluate the information submitted. Based on my inquiry of the person or persons who manage the system of those persons directly responsible for gathering the information, the information is, to the best of my knowledge and belief, true, accurate, and complete. I am aware that there is significant penalties for submitting false information, including the possibility of fine and imprisonment for knowing violations."

Name and Official Title **Jeffrey M. Dietlin** **Water Resources Supervisor**

Signature

Telephone Number **231-775-2841**

Date Signed **10/29/2003**

Upon request from the State, you may be required to submit additional information necessary to access biosolids use or disposal practices at your facility or to identify appropriate permitting requirements.

PLEASE RETURN COMPLETED FORMS TO:

BIOSOLIDS PROGRAM
TECHNICAL SUPPORT UNIT
WATER DIVISION
DEPARTMENT OF ENVIRONMENTAL QUALITY
PO BOX 30273
LANSING MI 48909-7773

EQP 5865 (7/2003)

2003 ANNUAL SLUDGE REPORT

Certification Statement
40CFR,503.17(a)(4)(ii)(A)

I certify, under penalty of law, that the management practices in 503.14, the site restrictions in 503.32(b)(5), and the vector attraction reduction requirements in 503.33(b)(9) have been met for each site on which bulk sewage sludge is applied. This determination has been made under my direction and supervision in accordance with the system designed to ensure that qualified personnel properly gather and evaluate the information used to determine that the management practices, site restrictions, and vector attraction reduction requirements have been met. I am aware that there are significant penalties for false certifitcation including the possibility of fine and imprisonment.

Jeffrey M. Dietlin,
Water Resources Supervisor
Cadillac Utilities Department

10/29/2003
Date

2003 ANNUAL SLUDGE REPORT

Certification Statement
40CFR,503.17(a)(4)(i)(B)

I certify under, penalty of law, that the Class B pathogen requirements in 503.32(b) and the vector attraction reduction requirement in 503.33(b)(1) have been met. This determination has been made under my direction and supervision in accordance with the system designed to ensure that qualified personnel properly gather and evaluate the information used to determine that the pathogen requirements and vector attraction reduction requirements have been met. I am aware that there are significant penalties for false certifitcation including the possibility of fine and imprisonment.

Jeffrey M. Dietlin
Water Resources Supervisor
Cadillac Utilities Department

10/29/2003
Date

BIOSOLIDS PRODUCTION - 2003

BIOPRO.03

OCTOBER 2002 INJECTION

Farm Site	Gallons	Solids Conc. %	Solids % Volatile	Pounds Applied	Tons Produced
GB001	432,000	5	45.6	180144.0	90.1
GB002	422,800	5	45.6	176307.6	88.2

MAY 2003 INJECTION

Farm Site	Gallons	Solids Conc. %	Solids % Volatile	Pounds Applied	Tons Produced
WE001	396,500	5.33	47.9	176253.0	88.1

VOLATILE SOLIDS REDUCTION - 2003

Month	Primary Sludge % Volatile	Injected % Volatile	Volatile Solid Reduction %
October	63.26	45.63	
November	64.51		
December	65.81		
January	67.48		
February	66.19		
March	65.72		
April	67.88		
May	65.80	47.94	$\frac{\text{In - Out}}{\text{In - (in x out)}}$
June	64.76		
July	64.30		
August	61.98		$\frac{0.65-0.47}{.65 - (.65 \times .47)}$
September	62.03		
Average	64.98	46.79	52.61% Reduction ✓

CITY OF CADILLAC WWTP - 2003
SLUDGE DIGESTION TEMPERATURE AND VOLUME - MONTHLY AVERAGE

MONTH	DIGESTER TEMP. DEG. F	PRI. SLUDGE PUMPED
October	98.10	414,000
November	94.30	332,400
December	92.00	415,600
January	92.45	467,200
Febuary	95.46	428,400
March	91.32	766,000
April	94.90	477,600
May	98.70	580,000
June	101.10	502,000
July	103.60	475,800
August	106.80	541,200
September	105.70	494,440
	Average 97.9	Total 5,894,640

Volume of 50' dia. primary digester is 352,310 gallons - top 2'for cover = 323,000 gallons
2002 Daily Sludge Production = 16150 gallons
Sludge Retention Time = 20.0 days

Sludge Storage Capacity =	Secondary Digester	323,000
	#3 Storage Tank	577,000
	#4 Storage Tank	691,100

Total sludge treatment volume is 1,914,000 gallons.

LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : ***024988***

Date Reported : ***09/26/02***

Project Desc. : ***Analysis of one sludge sample.***

Sample ID : ***"Tank #4 Biosolids"***

Sampled By : *JDC of City of Cadillac WWTP*

Sample Date : *9/11/02*

Sample Time : *1130*

Date Received : *9/12/02*

Sample Type : *sludge*

KAR Sample No. : ***024988-01***

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, Hg	Completed		EPA 7471A	09/19/02	BLF	
Prep, metals	Completed		EPA 3050	09/18/02	BLF	
Aluminum, total	6180	mg/kg total solids	EPA 6010B	09/26/02	KLZ	Detection Limit = 66
Arsenic, total	20.3	mg/kg total solids	EPA 6020	09/23/02	DBL	Detection Limit = 3.3
Cadmium, total	3.5	mg/kg total solids	EPA 6020	09/23/02	DBL	Detection Limit = 0.7
Calcium, total	27,500	mg/kg total solids	EPA 6010B	09/26/02	KLZ	Detection Limit = 1300
Chromium, total	109	mg/kg total solids	EPA 6020	09/23/02	DBL	Detection Limit = 3.3
Copper, total	903	mg/kg total solids	EPA 6020	09/23/02	DBL	Detection Limit = 15
Lead, total, by ICP	82	mg/kg total solids	EPA 6020	09/23/02	DBL	Detection Limit = 3.3
Magnesium, total	4300	mg/kg total solids	EPA 6010B	09/26/02	KLZ	Detection Limit = 66
Mercury, total	6.0	mg/kg total solids	EPA 7471A	09/20/02	KLZ	Detection Limit = 1.7
Molybdenum, total	37	mg/kg total solids	EPA 6020	09/23/02	DBL	Detection Limit = 3.3
Nickel, total	31	mg/kg total solids	EPA 6020	09/23/02	DBL	Detection Limit = 3.3
Potassium, total	2160	mg/kg total solids	EPA 6010B	09/26/02	KLZ	Detection Limit = 66
Selenium, total	4.3	mg/kg total solids	EPA 6020	09/23/02	DBL	Detection Limit = 3.3
Silver, total	56.4	mg/kg total solids	EPA 6020	09/23/02	DBL	Detection Limit = 3.3
Sodium, total	2860	mg/kg total solids	EPA 6010B	09/26/02	KLZ	Detection Limit = 66
Zinc, total	1430	mg/kg total solids	EPA 6020	09/23/02	DBL	Detection Limit = 15
Chloride	2810	mg/kg total solids	EPA 300.0A	09/17/02	MTW	Detection Limit = 500
Nitrogen, ammonia	19,600	mg/kg total solids	EPA 350.1	09/13/02	MTW	Detection Limit = 400
Nitrogen, nitrate	<20	mg/kg total solids	EPA 353.2	09/13/02	KLT	Detection Limit = 20
Nitrogen, total kjeldahl	44,300	mg/kg total solids	EPA 351.2	09/17/02	MTW	Detection Limit = 2000
Phosphorus, total (as P)	43,400	mg/kg total solids	EPA 365.2	09/18/02	DME	Detection Limit = 800
Solids, total	5.00	% by weight	SM(18) 2540 B	09/13/02	MTW	Detection Limit = 0.01
Solids, volatile	45.63	% of total solids	EPA 160.4	09/13/02	MTW	Detection Limit = 0.01
Sulfate	2260	mg/kg total solids	EPA 300.0A	09/17/02	MTW	Detection Limit = 1000
Waste pH	7.8	S.U.	EPA 9045C	09/17/02	KLT	

HOOSIER MICROBIOLOGICAL HML LABORATORY

Testing • Research • Consulting

Multiple Analysis Report

Sample: 82720
October 28, 2003

Mr. Jeff Dietlin
City of Cadillac
200 N. Lake St.
Cadillac, MI 49601

RE: PWS ID#: Unavailable
City of Cadillac
Sludge tank #4
1121 Plett Rd.
Cadillac, MI 49601

Dear Mr. Dietlin:

The following are the result(s) of the test(s) performed on the sample(s) received at HML, Inc. at 10:15 AM, 09/05/2002, and collected at 1:30 PM, 09/03/2002:

TEST - METHOD	RESULT	MDL*	Date Complete
Fecal Coliform-SM9221E (MPN)	10 MPN/g dry	4.6 MPN/g dry	09/09/2002
Enteric Virus-ASTM D4994-89	<1 pfu/4g dry	1 pfu/4g dry	10/18/2002
Solids, total-EPA160.3	4.6%	0.01%	09/06/2002
Salmonella-EPA 625/R-92/013	<3 MPN/4g dry	3 MPN/4 g dry	09/07/2002
Helminth Ova-EPA/625/R-92/013	<1/4g dry	1/4 g dry	10/01/2002

*Minimum Detection Level

This testing was completed by K.B. Please feel free to contact us if we can be of further service to you

Sincerely,

Donald A Hendrickson / SKP

Donald A. Hendrickson, Ph.D.
President - Microbiologist
Chemistry Lab #C-18-01
Microbiological Lab #M-18-03

DAH/skp

912 West McGalliard, Muncie, IN 47303-1702 Phone: (765) 288-1124 Fax:(765) 288-8378
E-mail: 102255.152@compuserve.com Web Site: http://www.hml.com

LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : *032095*

Date Reported : *05/02/03*

Project Desc. : *Analysis of one sludge sample.*

Sample ID : *"Biosolids"*

Sampled By :

Sample Date : *4/24/03*

Sample Time : *1:00pm*

Date Received : *4/25/03*

Sample Type : *sludge*

KAR Sample No. : *032095-01*

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, Hg	Completed		EPA 7471A	04/29/03	MDH	
Prep, metals	Completed		EPA 3050	04/28/03	MDH	
Aluminum, total	6110	mg/kg total solids	EPA 6010B	05/02/03	DBL	Detection Limit = 30
Arsenic, total	21	mg/kg total solids	EPA 6010B	04/29/03	PML	Detection Limit = 4
Cadmium, total	2.1	mg/kg total solids	EPA 6010B	04/29/03	PML	Detection Limit = 1
Calcium, total	25,300	mg/kg total solids	EPA 6010B	05/02/03	DBL	Detection Limit = 60
Chromium, total	114	mg/kg total solids	EPA 6010B	04/29/03	PML	Detection Limit = 8
Copper, total	925	mg/kg total solids	EPA 6010B	04/29/03	PML	Detection Limit = 8
Lead, total, by ICP	61	mg/kg total solids	EPA 6010B	04/29/03	PML	Detection Limit = 8
Magnesium, total	3650	mg/kg total solids	EPA 6010B	05/02/03	DBL	Detection Limit = 60
Mercury, total	2.59	mg/kg total solids	EPA 7471A	04/30/03	DBL	Detection Limit = 0.05
Molybdenum, total	37	mg/kg total solids	EPA 6010B	05/02/03	DBL	Detection Limit = 10
Nickel, total	28	mg/kg total solids	EPA 6010B	05/02/03	DBL	Detection Limit = 10
Potassium, total	1510	mg/kg total solids	EPA 6010B	05/02/03	DBL	Detection Limit = 60
Selenium, total	<10	mg/kg total solids	EPA 6010B	04/29/03	PML	Elevated detection limit due to low solids content. Detection Limit = 10
Silver, total	38	mg/kg total solids	EPA 6010B	04/29/03	PML	Detection Limit = 4
Sodium, total	2320	mg/kg total solids	EPA 6010B	05/02/03	DBL	Detection Limit = 60
Zinc, total	1700	mg/kg total solids	EPA 6010B	04/29/03	PML	Detection Limit = 8
Chloride	3110	mg/kg total solids	SM(18) 4500-Cl-E	04/28/03	MTW	Detection Limit = 94
Nitrogen, ammonia	18,600	mg/kg total solids	EPA 350.1	04/28/03	MTW	Detection Limit = 376
Nitrogen, nitrate	<18.8	mg/kg total solids	EPA 353.2	04/25/03	MTW	Detection Limit = 18.8
Nitrogen, total kjeldahl	49,400	mg/kg total solids	EPA 351.2	04/29/03	DMC	Detection Limit = 1880
Phosphorus, total (as P)	25,000	mg/kg total solids	EPA 365.2	04/29/03	BLF	Detection Limit = 751
Solids, total	5.33	% by weight	SM(18) 2540 B	04/28/03	MDH	Detection Limit = 0.01
Solids, volatile	47.94	% of total solids	EPA 160.4	04/28/03	MDH	Detection Limit = 0.01
Sulfate	477	mg/kg total solids	EPA 300.0A	04/28/03	MTW	Detection Limit = 188
Waste pH	7.7	S.U.	EPA 9045C	05/01/03	KLT	

HOOSIER MICROBIOLOGICAL

HML

LABORATORY

Testing • Research • Consulting

Multiple Analysis Report

Sample: 94835

July 9, 2003

Mr. Jeff Dietlin
City of Cadillac
200 N. Lake St.
Cadillac, MI 49601

RE: PWS ID#: Unavailable
City of Cadillac
Unavailable
1121 Plett Rd.
Cadillac, MI 49601

Dear Mr. Dietlin:

The following are the result(s) of the test(s) performed on the sample(s) received at HML, Inc. at 10:50 AM, 05/22/2003, and collected at 12:00 PM, 05/10/2003:

TEST - METHOD	RESULT	MDL*	Date Complete
Fecal Coliform-SM9221E (MPN)	24 MPN/g dry	4 MPN/g dry	05/26/2003
Enteric Virus-ASTM D4994-89	<1 pfu/4g dry	1 pfu/4g dry	07/08/2003
Solids, total-EPA160.3	4.5%	0.01%	05/23/2003
Salmonella-EPA 625/R-92/013	<3 MPN/4g dry	3 MPN/4 g dry	05/28/2003
Helminth Ova-EPA/625/R-92/013	<1/4g dry	1/4 g dry	06/25/2003

*Minimum Detection Level

This testing was completed by K.N. Please feel free to contact us if we can be of further service to you.

Sincerely,

Donald A. Hendrickson, Ph.D.
President - Microbiologist
Chemistry Lab #C-18-01
Microbiological Lab #M-18-03

DAH/skp

*Exceeds Holding Time

PLANT SUMMARY

Cadillac

Year Summary beginning October, 2002

	PlntEffi 50050 FLOW MGD Ave	PlntEffi 50050 FLOW MGD Sum	MISC M94008 POWER KILOWATTS Sum	MISC FUEL AUX. FUEL Sum	CL2Pro 50059 CL2 FEED LB Sum	CL2Pro SO2 Sulfer Dioxide Sum	SecPro M91010 FERRIC CHLORD. LB Sum	RawSldge 00056 FLOW GPD Sum	Supernat 00056 FLOW GPD Sum	Digester 00011 TEMP WATER F Average
Oct	2.084	64.601	130.20	1605.0	956.88	403.00	24585	414000	242480	98.06
Nov	1.990	59.685	117.60	3802.0	908.15	482.00	22898	332400	265430	94.30
Dec	1.881	58.301	124.20	5353.0	832.84	475.00	21109	415600	302560	92.00
Jan	2.011	62.340	126.80	5452.0	784.11	543.00	22263	467200	333060	92.45
Feb	2.018	56.502	114.20	5663.0	549.32	438.00	23117	428400	287920	95.46
Mar	2.030	62.937	122.60	4935.0	562.61	423.00	23121	766000	40260	91.32
Apr	2.002	60.066	117.40	4306.0	801.83	432.00	21449	477600	322080	94.90
May	2.047	63.459	117.40	3648.0	655.64	422.00	25164	580000	286700	98.74
Jun	2.094	62.815	112.80	1584.0	1027.76	478.00	24087	502000	259000	101.07
Jul	2.288	70.935	120.20	1753.0	1156.23	424.00	26882	475800	350140	103.61
Aug	2.346	72.724	122.60	1612.0	1151.80	496.00	28376	541200	335500	106.84
Sep	2.232	66.953	114.80	1668.0	1138.51	405.00	24930	494440	269620	105.73
Sum	25.022	761.316	1440.80	41381.0	10525.68	5421.00	287980	5894640	3294750	1174.50
Avg	2.085	63.443	120.07	3448.4	877.14	451.75	23998	491220	274563	97.87
Max	2.346	72.724	130.20	5663.0	1156.23	543.00	28376	766000	350140	106.84
Max3	2.289	70.204	124.00	5489.3	1148.85	500.00	26729	607867	318420	105.40
Min	1.881	56.502	112.80	1584.0	549.32	403.00	21109	332400	40260	91.32

Year Summary beginning October, 2002

	Raw 00310 BOD 5 DAY Average	Raw 00530 RESIDUE TOT NFLT Average	Raw 00665 PHOS TOTAL P WET Average	Final 00310 BOD 5 DAY Average	Final 00530 RESIDUE TOT NFLT Average	Final 00665 PHOS TOTAL P WET Average	RawSldge 70322 SLUDGE SOLIDS			
Oct	150.5	137.3	3.814	2.504	1.409	0.180	63.261			
Nov	153.1	141.4	3.652	2.794	1.271	0.169	64.506			
Dec	174.0	147.4	4.814	3.905	3.784	0.211	65.806			
Jan	167.2	139.3	4.163	4.398	4.027	0.260	67.482			
Feb	166.1	155.6	4.308	3.685	3.600	0.203	66.190			
Mar	156.6	137.8	3.869	3.437	2.164	0.160	65.720			
Apr	168.5	145.7	3.430	3.848	4.436	0.257	67.882			
May	158.0	150.4	3.365	2.560	1.891	0.196	65.798			
Jun	152.9	134.7	4.151	1.950	1.509	0.190	64.757			
Jul	167.4	143.5	3.897	2.706	2.609	0.198	64.303			
Aug	135.9	139.4	3.156	1.859	1.860	0.203	61.979			
Sep	132.5	128.9	4.024	2.079	1.364	0.187	62.030			
Sum	1882.7	1701.3	46.645	35.725	29.923	2.414	779.714			
Avg	156.9	141.8	3.887	2.977	2.494	0.201	64.976			
Max	174.0	155.6	4.814	4.398	4.436	0.260	67.882			
Max3	169.1	147.4	4.428	3.996	3.804	0.225	66.597			
Min	132.5	128.9	3.156	1.859	1.271	0.160	61.979			

YEAR END REPORT

Cadillac

Year Summary beginning October, 2002

	Raw 00310 BOD 5 DAY Average	Final 00310 BOD 5 DAY Average	PlntEffi ~P00310 % Removal Average	Raw 00530 RESIDUE TOT NFLT Average	Final 00530 RESIDUE TOT NFLT Average	PlntEffi ~P00530 % Removal Average	Raw 00610 NH3 N AMMONIA Average	Final 00610 NH3 N AMMONIA Average	PlntEffi ~P00610 % Removal Average	
Oct	150.517	2.504	98.333	137.304	1.409	98.990	22.752	0.251	98.863	
Nov	153.088	2.794	98.160	141.353	1.271	99.065	24.971	8.131	64.906	
Dec	173.989	3.905	97.692	147.368	3.784	97.347	25.549	0.593	97.804	
Jan	167.205	4.398	97.300	139.273	4.027	97.177	24.086	0.593	97.558	
Feb	166.112	3.685	97.723	155.579	3.600	97.475	25.237	2.906	89.099	
Mar	156.618	3.437	97.804	137.818	2.164	98.345	16.470	1.469	91.593	
Apr	168.540	3.848	97.642	145.714	4.436	96.947	21.010	0.808	96.535	
May	157.994	2.560	98.356	150.400	1.891	98.737	25.130	0.330	98.601	
Jun	152.868	1.950	98.666	134.727	1.509	98.825	23.755	0.279	98.866	
Jul	167.391	2.706	98.336	143.455	2.609	98.068	24.041	0.370	98.427	
Aug	135.918	1.859	98.623	139.400	1.860	98.594	23.360	0.422	98.126	
Sep	132.458	2.079	98.402	128.909	1.364	98.924	23.282	0.718	97.456	
Sum	1882.698	35.725	1177.037	1701.301	29.923	1178.494	279.643	16.870	1127.836	
Avg	156.892	2.977	98.086	141.775	2.494	98.208	23.304	1.406	93.986	
Max	173.989	4.398	98.666	155.579	4.436	99.065	25.549	8.131	98.866	
Max3	169.102	3.996	98.542	147.407	3.804	98.543	24.957	3.106	98.632	
Min	132.458	1.859	97.300	128.909	1.271	96.947	16.470	0.251	64.906	

YEAR END 2

Cadillac

Year Summary beginning October, 2002

	Raw 00665 PHOS TOTAL P WET Average	Final 00665 PHOS TOTAL P WET Average	PlntEffi ~P00665 % Removal Average		RawSldge 70318 SLUDGE SOLIDS		Supernat 70318 SLUDGE SOLIDS			
Oct	3.814	0.180	95.148		3.501		0.184			
Nov	3.652	0.169	95.052		3.817		0.187			
Dec	4.814	0.211	95.530		3.699		0.501			
Jan	4.163	0.260	93.596		3.399		0.564			
Feb	4.308	0.203	95.035		3.382		0.428			
Mar	3.869	0.160	95.730		3.761		0.590			
Apr	3.430	0.257	91.447		3.714		0.256			
May	3.365	0.196	92.835		2.896		0.238			
Jun	4.151	0.190	95.398		3.004		0.234			
Jul	3.897	0.198	94.831		3.115		0.194			
Aug	3.156	0.203	93.491		3.028		0.226			
Sep	4.024	0.187	95.317		3.170		0.223			
Sum	46.645	2.414	1133.409		40.485		3.825			
Avg	3.887	0.201	94.451		3.374		0.319			
Max	4.814	0.260	95.730		3.817		0.590			
Max3	4.428	0.225	95.243		3.672		0.527			
Min	3.156	0.160	91.447		2.896		0.184			

State of Michigan
Department of Natural Resources

WASTE DISPOSAL REPORT

October, 19 2002
(Month)

Cadillac WWTP
(Facility)

(Superintendent's Signature)

The filing of this report may be required under the authority of Act 98 P.A. 1913, as amended, being Section 325.208. If so required, failure to submit required reports shall be subject to penalty as set forth in Section 325.213.

Use of this form is encouraged as an operational control record, where applicable, when filing of such report is not required.

MDNR Site # 28001

of seasons utilized to date 1
Acres used this month 36
Total acres in site 40

WASTE APPLIED				WASTE ANALYSIS AND SOIL LOADING RATES																					
				Nitrogen				Total Phosphorus TP		Potassium K		Lead Pb		Zinc Zn		Copper Cu		Nickel Ni		Cadmium Cd		Chromium Cr			
Gallons or Cubic Yards	% Solids	% VS	Dry Tons per ac	TKN %	NH_4 %	NO_3 %	AVAN lb/ac	%	lb/ac	%	lb/ac	mg/kg	lb/ac	mg/kg	lb/ac	mg/kg	lb/ac	mg/kg	lb/ac	mg/kg	lb/ac	mg/kg	lb/ac	mg/kg	lb/ac
432,000	5.0	45.6		4.4	2.0	<.01		4.3		.22		82		1430		903		31		3.5		109			
432000			2.5				124		215		11		.41		7.15		4.5		.155		.017		.54		
432000			2.5				124		215		11		.41		7.15		4.5		.156		.017		.54		

CROP AND SOIL DATA

Crop to be fertilized Corn

Subsequent Crop Corn

CEC 5.7 meq/1

pH 5.3

Bray P1 120

K 70

Crop Yield Goal 20 T/a

NITROGEN RECOMMENDE

N ______ lb

Acceptable Metal Accumulat

	Total	Yearly
Pb	570	28.5
Zn	285	142.5
Cu	142.5	7.1
Ni	57	2.8
Cd	4.5	0.45

Cd 0.45 lb/ac/yr (0.5 kg/ha/y

REMARKS:

Date of Waste Analysis

PR
Rev

October, 19 2002
(Month)
Cadillac WWTP
(Facility)

(Superintendent's Signature)

State of Michigan
Department of Natural Resources

WASTE DISPOSAL REPORT

The filing of this report may be required under the authority of Act 98 P.A. 1913, as amended, being Section 325.208. If so required, failure to submit required reports shall be subject to penalty as set forth in Section 325.213.

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MDNR Site # 28002
# of seasons utilized to date ______
Acres used this month 32
Total acres in site 40

WASTE APPLIED				WASTE ANALYSIS AND SOIL LOADING RATES																					
				Nitrogen				Total Phosphorus TP		Potassium K		Lead Pb		Zinc Zn		Copper Cu		Nickel Ni		Cadmium Cd		Chromium Cr			
Gallons or Cubic Yards	% Solids	% VS	Dry Tons per ac	TKN %	NH_4 %	NO_3 %	AVAN lb/ac	%	lb/ac	%	lb/ac	mg/kg	lb/ac	mg/kg	lb/ac	mg/kg	lb/ac	mg/kg	lb/ac	mg/kg	lb/ac	mg/kg	lb/ac	mg/kg	lb/ac
422,800	5.0	45.6		4.1	20	[illegible]		4.3		.22		82		1436		963		31		3.5		109			
422,800			2.45				121.5		211		10.8		.40		7.00		4.4		.151		.017		.53		
422,800			2.45				121.5		211		10.8		.40		7.00		4.4		.151		.017		.53		
													.40		7.00		4.4		.151		.017		.53		

CROP AND SOIL DATA

Crop to be fertilized C[illegible]
Subsequent Crop Corn
CEC 5.3 meq/
pH 5.0
Bray P1 95
K 105
Crop Yield Goal 20 T/a
NITROGEN RECOMMEND
N ______
Acceptable Metal Accumul

	Total	Yearly
Pb	530	26.5
Zn	265	13.2
Cu	132.5	6.6
Ni	53	2.65
Cd	4.5	0.45

Cd 0.45 lb/ac/yr (0.5 kg/ha

REMARKS:

Date of Waste Analysis

PF
Re

MAY (Month), 19 ~~19~~ 2003

Cadillac WWTP (Facility)

(Superintendent's Signature)

State of Michigan
Department of Natural Resources

WASTE DISPOSAL REPORT

The filing of this report may be required under the authority of Act 98 P.A. 1913, as amended, being Section 325.208. If so required, failure to submit required reports shall be subject to penalty as set forth in Section 325.213.

Use of this form is encouraged as an operational control record, where applicable, when filing of such report is not required.

MDNR Site # WE 001

\# of seasons utilized to date 2

Acres used this month 40

Total acres in site 40

WASTE APPLIED				WASTE ANALYSIS AND SOIL LOADING RATES																					
Gallons or Cubic Yards	% Solids	% VS	Dry Tons per ac	Nitrogen TKN %	Nitrogen NH_4 %	Nitrogen NO_3 %	Nitrogen AVAN lb/ac	Total Phosphorus TP %	Total Phosphorus TP lb/ac	Potassium K %	Potassium K lb/ac	Lead Pb mg/kg	Lead Pb lb/ac	Zinc Zn mg/kg	Zinc Zn lb/ac	Copper Cu mg/kg	Copper Cu lb/ac	Nickel Ni mg/kg	Nickel Ni lb/ac	Cadmium Cd mg/kg	Cadmium Cd lb/ac	Chromium Cr mg/kg	Chromium Cr lb/ac	mg/kg	lb/ac
396,500	5.33	47.9		4.9	1.9	<0.01		2.5		.15		61		1700		925		28		21		114			
396500			2.2				110		110		6.6		.27		7.48		4.1		.12		.009		0.5		
396,500			2.2				110		110		6.6		.27		7.48		4.1		.12		.009		0.5		

CROP AND SOIL DATA

Crop to be fertilized Corn

Subsequent Crop Corn

CEC 8.4 meq/1(

pH 5.18 s

Bray P1 18 p

K 58 p

Crop Yield Goal 20.74

NITROGEN RECOMMENDE(

N ____ lb

Acceptable Metal Accumulati

	Total	Yearly	
Pb	840	42	lb
Zn	420	21	lb
Cu	210	10.5	lb
Ni	84	4.2	lb
Cd	4.5	0.45	lb
			lb
			lb

Cd 0.45 lb/ac/yr (0.5 kg/ha/yr

REMARKS:

Date of Waste Analysis

PR 55
Rev 1

Class B Site Restrictions
R 323.2414(3) f

All of the following provisions apply to site restrictions:

(i) A landowner shall not harvest food crops that have harvested parts which touch the biosolids/soil mixture and which are totally above the land surface for 14 months after biosolids are applied.

(ii) A landowner shall not harvest food crops that have harvested parts below the surface of the land for 20 months after biosolids are applied if the biosolids remain on the land surface for 4 months or longer before incorporation into the soil.

(iii) A landowner shall not harvest food crops that have harvested parts below the surface of the land for 38 months after biosolids are applied if the biosolids remain on the land surface for less than 4 months before incorporation into the soil.

(iv) A landowner shall not harvest food crops, feed crops, and fiber crops for 30 days after biosolids are applied.

(v) A landowner shall not graze animals on the land for 30 days after biosolids are applied.

(vi) A landowner shall not harvest turf grown on land where biosolids are applied for 1 year after biosolids are applied if the harvested turf is placed on either land that has a high potential for public exposure or a lawn, unless otherwise specified by the permitting authority.

(vii) A landowner shall restrict public access to land that has a high potential for public exposure of r1 year after biosolids are applied.

(viii) A landowner shall restrict public access to land with a low potential for public exposure for 30 days after biosolids are applied.

Certification

"I certify, under penalty of law, that the information that will be used to determine compliance with the site restrictions in R 323.2414(3)(f) has been prepared under my direction and supervision in accordance with the system designed to ensure that qualified personnel properly gather and evaluate the information. I am aware that there are significant penalties for false certification including the possibility of fine and imprisonment.

Project: Cadillac Wastewater Treatment Plant

Signature: ______________ Date: 9-20-03

Kevin Bonney
BioTech Agronomics, Inc.

R 323.2410 Management Practices

Rule 2410

(1) A person shall not apply bulk biosolids to the land if it is likely to adversely affect a threatened or endangered species listed under section 36503 of the act or its designated critical habitat.

(2) A person shall not apply bulk biosolids to agricultural land, a forest, a public contact site, or a reclamation site that is flooded, saturated with water, frozen, or snow-covered so that the bulk biosolids enter a wetland or other waters of the state.

(3) A person may subsurface inject biosolids on frozen or snow-covered ground as long as there is substantial soil coverage of the applied biosolids. A person shall not surface apply bulk biosolids, other than exceptional quality biosolids, on frozen or snow-covered ground, unless otherwise approved by the department.

(4) A person shall not apply bulk biosolids on lands having a slope o more than 6% for surface application or more than 12% for subsurface injected biosolids, unless the person uses the bulk biosolids in accordance with a department-approved site management plan.

(5) A person shall apply bulk biosolids to agricultural land, a forest, a public contact site, or a reclamation site at an application rate that is equal to, or less than, the agronomic rate, unless the person that applies bulk biosolids in accordance with a department-approved site management plan.

(6) A generator or distributor shall affix a label to the bag or other container in which biosolids are sold or given away for application to the land or the generator or distributor shall provide an information sheet to the person who receives biosolids sold or given away in another container for application to the land. The label or information sheet shall contain all of the following information:

    (a) The name and address of the person who prepared the biosolids that are sold or given away in a bag or other container for application to the land.

    (b) A statement that the application of the biosolids to the land is prohibited unless applied according to the instructions on the label or information sheet.

    (c) The annual whole biosolids application rate for biosolids that do not cause any of the annual pollutant loading rates in table 4 of R 323.2409(5)(d) to be exceeded.

(7) A person that applies biosolids shall perform soil fertility tests on soils sampled from each application site before initial biosolids application. The person shall resample and test on a regular basis so that the last soil fertility test is not more than 2 years old at the time of the next biosolids application.

(8) For agricultural land, a person shall apply biosolids in accordance with agronomic rates. If the Bray P1 soil test level exceeds 300 pounds (P) per acre (150 ppm), or if the Mehlich 3 soil test level exceeds 340 pounds (P) per acre (170 ppm) in site soils, then the person shall not apply biosolids until the soil P test level decreases to less than 1 of these values.

Certification

"I certify, under penalty of law, that the information that will be used to determine compliance with the management practices in R 323.2410 have been prepared for each site on which bulk biosolids are applied under my direction and supervision in accordance with the system designed to ensure that qualified personnel properly gather and evaluate the information. I am aware that there are significant penalties for false certification including the possibility of fine and imprisonment."

Project: Cadillac Wastewater Treatment Plant

Signature: K B Date: 9-20-03

Kevin Bonney
BioTech Agronomics, Inc.

Vector Attraction Requirement
R 323.2415(4), i, j

(i) All of the following provisions apply to biosolids that are injected:
   (i) A person who applies biosolids shall inject below the surface of the land.
   (ii) A person who applies biosolids shall ensure that a significant amount of the biosolids is not present on the land surface within 1 hour after the biosolids are injected.
   (iii) If the biosolids that are injected below the surface of the land are class A with respect to pathogens, then a person who applies the biosolids shall ensure that the biosolids are injected below the land surface within 8 hours after the biosolids have been discharged from the pathogen treatment process.

(j) A person who applies biosolids to the land surface shall ensure that the biosolids are incorporated into the soil within 6 hours after application to or placement on the land, unless otherwise specified by the permitting authority. If biosolids that are incorporated into the soil are class A with respect to pathogens, then the person shall apply the biosolids to, or place the biosolids on, the land within 8 hour after the biosolids have been discharged from the pathogen treatment process.

Certification

"I certify, under penalty of law, that the information that will be used to determine compliance with the vector attraction reduction requirement in either R 323.2415(4)(i) or (j) has been prepared under my direction and supervision in accordance with the system designed to ensure that qualified personnel properly gather and evaluate the information. I am aware that there are significant penalties for false certification including the possibility of fine and imprisonment.

Project: Cadillac Wastewater Treatment Plant

Signature: _______________ Date: 9-26-03
Kevin Bonney
BioTech Agronomics, Inc.

Pathogens and Vector Attraction Reduction

Certification

"I certify, under the penalty of law, that information that will be used to determine compliance with the pathogen requirements in R323.2414 (3) has been prepared under my direction and supervision in accordance with the system designed to ensure that qualified personnel properly gather and evaluate the information. I am aware that there are significant penalties for false certification including the possibility of fine and imprisonment."

"I certify, under penalty of law, that the information that will be used to determine compliance with the management practices in R323.2410 and the vector attraction reduction requirement in R323.2415 (4)(i) has been prepared under my direction and supervision in accordance with the system designed to ensure that qualified personnel properly gather and evaluate the information. I am aware that there are significant penalties for false certification including the possibility of fine and imprisonment."

Vector attraction reduction requirements for R323.2415 (4)(i) are met through:

- (i) A person who applies biosolids shall inject below the surface of the land.
- (ii) A person who applies biosolids shall ensure that a significant amount of the biosolids is not present on the land surface within one hour after the biosolids are injected.
- (iii) A person who applies biosolids to the land surface shall ensure that the biosolids are incorporated into the soil within 6 hours after application to or placement on the land, unless otherwise specified by the permitting authority.

Generator: Cadillac Wastewater Treatment Plant

____________________________ Date: ________________